# Diagnosis: Colorectal cancer

## Treatement 1: Crizotinib

### Chosen clinical trials:

1. 'MEK and MET Inhibition in Colorectal Cancer' completed phase1 https://clinicaltrials.gov/study/NCT02510001

   Population: 30

   Time: Dose Expansion phase: change from baseline and up to 12 months.

   Stable Disease: 7

   Progressive Disease: 22

   Early death from malignant disease: 1

### Chosen scientific papers:

Crizotinib blocked the HGF/STAT3/SOX13/c-MET axis, significantly inhibiting SOX13-mediated CRC migration, invasion and metastasis [1]. The dual ALK-MET inhibitor crizotinib plus vemurafenib showed rapid and marked effectiveness in treating BRAF-mutated colorectal cancer [2]. Crizotinib showed a partial response in one colorectal cancer patient [3]. Crizotinib treatment overcame cetuximab resistance in in vitro and in vivo studies [4]. Crizotinib was used in combination with tislelizumab and regorafenib, achieving a good clinical therapeutic effect without severe side effects [5]. Crizotinib has demonstrated enhanced cytotoxicity in ARID1A-deficient colorectal cancer cells [6]. Crizotinib improves the efficacy of treatment in colon cancer models when combined with EpAb2-6 [7]. TH588 and (S)-crizotinib have clear and distinct anti-tumour effects in 3D colorectal cancer cultures, but these effects occur through ROS-independent mechanisms [8]. Crizotinib exhibits moderate cytotoxicity on Caco-2 cell line [9].

[1] SOX13 promotes colorectal cancer metastasis by transactivating SNAI2 and c-MET. https://pubmed.ncbi.nlm.nih.gov/32111984

[2] MET-Driven Resistance to Dual EGFR and BRAF Blockade May Be Overcome by Switching from EGFR to MET Inhibition in BRAF-Mutated Colorectal Cancer. https://pubmed.ncbi.nlm.nih.gov/27325282

[3] ALK fusions in the pan-cancer setting: another tumor-agnostic target? https://pubmed.ncbi.nlm.nih.gov/37773318

[4] Inhibition of autocrine HGF maturation overcomes cetuximab resistance in colorectal cancer. https://pubmed.ncbi.nlm.nih.gov/38212428

[5] Partial response to crizotinib + regorafenib + PD-1 inhibitor in a metastatic https://pubmed.ncbi.nlm.nih.gov/38711893

[6] ARID1A loss enhances sensitivity to c-MET inhibition by dual targeting of GPX4 and iron homeostasis, inducing ferroptosis. https://pubmed.ncbi.nlm.nih.gov/40369167

[7] Epithelial cell adhesion molecule (EpCAM) regulates HGFR signaling to promote colon cancer progression and metastasis. https://pubmed.ncbi.nlm.nih.gov/37543570

[8] Differential anti-tumour effects of MTH1 inhibitors in patient-derived 3D colorectal cancer cultures. https://pubmed.ncbi.nlm.nih.gov/30692572

[9] In silico-in vitro discovery of untargeted kinase-inhibitor interactions from kinase-targeted therapies: A case study on the cancer MAPK signaling pathway. https://pubmed.ncbi.nlm.nih.gov/29803964